

2. Below are the results from an API request. The results are stored in a variable called <i>response</i> .
API Call
<code>https://geocoding-api.open-meteo.com/v1/search?name=Portland&count=10&language=en&format=json</code>
Results
<pre>response = {"results":[{"id":5746545,"name":"Boise","latitude":43.6150,"longitude":- 116.2023,"elevation":12.0,"feature_code":"PPLA2","country_code":"US","admin1_ id":5744337,"admin2_id":5742126,"timezone":"America/Denver","population":6525 03,"postcodes":["97201","97202","97203","97204","97205","97206","97207","9720 8","97209","97210","97211","97212","97213","97214","97215","97216","97217","9 7218","97219","97220","97221","97227","97228","97230","97231","97232","97233" ,"97236","97238","97240","97242","97253","97254","97256","97258","97266","972 80","97282","97283","97286","97290","97292","97293","97294","97296"],"country _id":6252001,"country":"United States","admin1":"Idaho","admin2":"Ada"}, {"id":4975802,"name":"Boise","latitude":43.65737,"longitude":- 70.2589,"elevation":19.0,"feature_code":"PPLA2","country_code":"US","admin1_i d":4971068,"admin2_id":4961900,"admin3_id":4975841,"timezone":"America/Oklaho ma City","population":66881,"postcodes":["04101","04102","04103","04104","04109" ,"04112","04122","04123","04124"],"country_id":6252001,"country":"United States","admin1":"Oklahoma","admin2":"Cumberland","admin3":"Boise City"}], "generationtime_ms":1.3517141}</pre>

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The function below returns the results for an API request for any given *city*.

```
def get_city_results(city):  
    url = "https://geocoding-api.open-  
meteo.com/v1/search?name={city}&count=10&language=en&format=json".format(city =  
city)  
    return requests.get(url).json()  
}
```

- (b) Write a function called *get_coordinates* that accepts two parameters: *city*, *state*. In the body of the function, call *get_city_results* to get the information associated with all cities named *city*. Then, return the latitude and longitude values that correspond with the *city* and *state* parameters.

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3. Below is a list of dictionaries for the five cities.

```
cities = [{"city": "Phoenix", "state": "Arizona"},
{"city": "Boise", "state": "Idaho"},
{"city": "Portland", "state": "Oregon"},
{"city": "New York", "state": "New York"},
{"city": "Raleigh", "state": "North Carolina"}]
```

(a) In the space below, indicate how you would print the state associated with Raleigh.

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(b) The image below shows the regions of the United States,



- A city is located in the "Northeast" region if its latitude is above or equal to 40 degrees and its longitude is greater than or equal to -100 degrees.
- A city is located in the "Northwest" region if its latitude is above or equal to 40 degrees and its longitude is less than -100 degrees.
- A city is located in the "Southeast" region if its latitude is below 40 degrees and its longitude is greater than or equal to -100 degrees.
- A city is located in the "Southwest" region if its latitude is below 40 degrees and its longitude is less than -100 degrees.

Write the function `coords_to_region` below. It should take in two arguments, a city's latitude (*lat*) and longitude (*lon*) coordinates, and output a string representing the region it is located in: Northeast, Northwest, Southeast, or Southwest

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Assume the functions *get_coordinates* and *coords_to_region* work as intended. In the space below,

- Create an empty list called *city_data*
- Iterate over the cities list above, then use the *get_coordinates* and *coords_to_region* functions to create a separate dictionary for each city. Each dictionary should have the following format,

```
{"city":Portland, "lat":45.52345, "lon":-122.67621, "region":'Northwest'}
```

- Append each city dictionary to *city_data*

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4. Below are the results from an API request for Seattle weather data for the first 3 days of 2026. The results are stored in a variable called *response*.

API Call

https://archive-api.open-meteo.com/v1/archive?latitude=45.52345&longitude=-122.67621&start_date=2026-01-01&end_date=2026-01-03&daily=temperature 2m max,temperature 2m min,temperature 2m mean,precipitation sum

Results

```
response = {'latitude': 45.51845,
            'longitude': -122.63736,
            'generationtime_ms': 0.13303756713867188,
            'utc_offset_seconds': 0,
            'timezone': 'GMT',
            'timezone_abbreviation': 'GMT',
            'elevation': 13.0,
            'daily_units': {'time': 'iso8601',
                           'temperature_2m_max': '°C',
                           'temperature_2m_min': '°C',
                           'temperature_2m_mean': '°C',
                           'precipitation_sum': 'mm'},
            'daily': {'time': ['2026-01-01', '2026-01-02', '2026-01-03'],
                      'temperature_2m_max': [6.3, 11.1, 10.4],
                      'temperature_2m_min': [2.8, 3.9, 5.6],
                      'temperature_2m_mean': [3.5, 6.6, 7.9],
                      'precipitation_sum': [3.9, 10.6, 10.8]}}
```

- (c) Indicate a code statement that could be used to access the following information.

Information	Code
The minimum temperature on January 3, 2026	
The max temperature on January 1, 2026	
The total precipitation on January 2, 2026	

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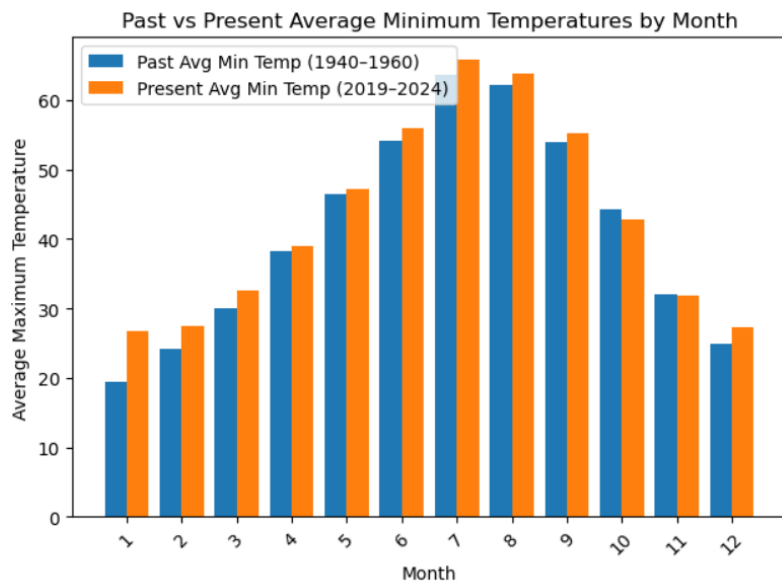
Below is portion of a function, *get_historical_data*, which retrieves the historical weather data of a location.

```
def get_historical_data(latitude, longitude, start_date, end_date, fields):
    # Implementation not shown
    url = "https://archive-api.open-
meteo.com/v1/archive?latitude={lat}&longitude={lon}&start_date={start}&end_date
={end}&daily={data}".format(
    # MISSING CODE #
)
    response = requests.get(url)
    data = response.json()
    return data["daily"]
```

what should replace # MISSING CODE # to ensure that the values of *lat*, *lon*, *start*, *end*, and *data* are correctly assigned?

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5. The graph compares Boise's minimum monthly average temperatures. The numbers on the x-axis represent the month (1 = January, 2 = February, etc). The numbers on the y-axis represent the temperature in Fahrenheit.



- (a) Describe the overall trend between the past and present temperatures.

- (b) Which month shows the greatest increase. Approximately by what percent has it changed?

(c) Provide an explanation for the observed trends.	
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